

Appendix 2: Lake Washington Analysis

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Here we show the GAM and DLM analysis for the Lake Washington dataset. For complete code, see the [full quarto document](#). We include some code snippets here for reference.

Data Processing

In Lake WA the major change ecosystem change is related to sewage in the lake – adding wastewater plants in the 1960s reduced phosphorous (Figure [S2.1](#)) and in turn, blue green algae (Figure [S2.2](#)) which can be seen by examining time series plots of the data.

We can then use the MARSS package to fit a DLM. We'll filter the data to only use counts before 1974, and log transform densities. We also z-score the covariate. This is a good example of imperfect data – there's a couple of missing observations in the covariate, which we can't have in a DLM or GAM – so let's just fill them in with a spline. We can do this as follows:

```
dat <- dplyr::filter(dat, Year < 1974) %>%
  dplyr::mutate(y = log(Bluegreens), zp = as.numeric(scale(TP)))

interpolated <- spline(dat$zp, xout = 1:nrow(dat))$y
dat$zp[which(is.na(dat$zp))] <- interpolated[which(is.na(dat$zp))]
```

The response data exhibit a strong seasonal cycle – and it's a good idea to de-season that to focus on the phosphorous effect. Here we de-season the data:

```
# Fill in a few missing values
dat$y_interp <- dat$y
interpolated <- spline(dat$y, xout = 1:nrow(dat))$y
missing <- which(is.na(dat$y_interp))
dat$y_interp[missing] <- interpolated[missing]
```

```
dat <- dplyr::group_by(dat, Month) %>%
  dplyr::mutate(month_mean = mean(y, na.rm=T)) %>%
  dplyr::ungroup() %>%
  dplyr::mutate(y_adj = y_interp - month_mean)
```

Last, we should look at the lags between driver and response relationships which we can do using a CCF (Figure S2.3). The CCF here indicates a 2-4 month lag, where TP and algae are more strongly associated with one another.

We'll use a lag of 4 months, which we can apply as follows:

```
lag_n <- 5
dat$lagged_zp <- c(rep(NA, lag_n), dat$zp[1:(nrow(dat)-lag_n)])
dat <- dat[-c(1:lag_n),]
```

DLMs

First, we need to define the model – this block is basically copied from the MARSS book, [the salmon survival case study](#)

```
m <- 2
TT <- nrow(dat)
B <- diag(m) ## 2x2; Identity
U <- matrix(0, nrow = m, ncol = 1) ## 2x1; both elements = 0
Q <- matrix(list(0), m, m) ## 2x2; all 0 for now
diag(Q)[1] <- 0.0000001
diag(Q)[2] <- c("q.beta")
#diag(Q) <- c("q.alpha", "q.beta") ## 2x2; diag = (q1,q2)
Z <- array(NA, c(1, m, TT)) ## NxMxT; empty for now
Z[1, 1, ] <- rep(1, TT) ## Nx1; 1's for intercept
Z[1, 2, ] <- dat$lagged_zp ## Nx1; predictor variable
A <- matrix(0) ## 1x1; scalar = 0
R <- matrix("r") ## 1x1; scalar = r
## only need starting values for regr parameters
inits_list <- list(x0 = matrix(c(0, 0), nrow = m))
## list of model matrices & vectors
mod_list <- list(B = B, U = U, Q = Q, Z = Z, A = A, R = R)
# convert response to matrix
dat_mat <- matrix(dat$y_adj, nrow = 1)
# fit the model -- crank up the maxit to ensure convergence
```

```
d1m_1 <- MARSS(dat_mat, inits = inits_list, model = mod_list,
control = list(maxit=4000))
```

Warning! Abstol convergence only. Maxit (=4000) reached before log-log convergence.

MARSS fit is

Estimation method: kem

Convergence test: conv.test.slope.tol = 0.5, abstol = 0.001

WARNING: Abstol convergence only no log-log convergence.

maxit (=4000) reached before log-log convergence.

The likelihood and params might not be at the ML values.

Try setting control\$maxit higher.

Log-likelihood: -163.6828

AIC: 335.3656 AICc: 335.6641

	Estimate
R.r	0.2872
Q.q.beta	0.2197
x0.X1	0.0131
x0.X2	1.9946

Initial states (x0) defined at t=0

Standard errors have not been calculated.

Use MARSSparamCIs to compute CIs and bias estimates.

Convergence warnings

Warning: the x0.X1 parameter value has not converged.

Type MARSSinfo("convergence") for more info on this warning.

Let's put the state estimates back into the original dataframe and do some plotting. The slope is generally positive, which is what we expect for the hypothesized relationship between bluegreen algae and total phosphorus – and exhibits clear variation through time (Figure S2.4). We can also plot the fitted values from our model (Figure S2.5).

And we can examine the residuals for a temporal trend, and see that because the DLM is flexible, there's not much evidence of a trend (Figure S2.6) and there are no issues with the q-q plot (Figure S2.7).

GAMs

We can fit a GAM model 5 ways, 1) so that the smoother is a function of phosphorus which represents a nonlinear functional relationship between total phosphorus and bluegreen algae

that is not temporally structured thus the relationship is stationary and fixed through time. This is how GAMs are typically structured for ecological analyses. 2) a temporally structured model such that the abundance of bluegreen algae varies as a function of time unrelated to phosphorus, 3) A temporally structured model so that the smooth function of phosphorus varies by time, and 4) a smooth on date as a driver of cyanobacteria where the smooth through time varies by phosphorus level. Finally, these relationships can be fit as 5) a tensor product which provide a multivariate function spanning the dimensions of time and phosphorus. We will not discuss tensor products in greater detail and only include them for completeness. We show the model predicted cyanobacteria trend through time using these five modeling approaches (Figure S2.8). While they are quite similar, the predictions diverge substantially at the end of the time series.

```
dat$n_date <- as.numeric(as.factor(dat$date))

gam_cr1 <- gam(y_adj ~ s(lagged_zp, bs = "cr"), data = dat) #1
gam_cr2 <- gam(y_adj ~ s(lagged_zp, by = n_date, bs = "cr"), data = dat) #2
gam_cr3 <- gam(y_adj ~ s(n_date, by = lagged_zp, bs = "cr"), data = dat) #3
gam_cr4 <- gam(y_adj ~ s(n_date), data = dat) #4
gam_cr5 <- gam(y_adj ~ te(lagged_zp, n_date), data = dat) #5
```

```
[1] "#26172AFF" "#414081FF" "#357BA2FF" "#3DB4ADFF" "#A8E1BCFF"
```

It is important to also examine the residuals from these models. For example, we can look at the residuals for the GAMs fit to the Lake Washington data. In looking across models, those that do not include time-varying parameters show evidence of a persistent trend in the residuals that is not accounted for by the model ((Figure S2.9) despite the q-q plot being reasonable for all model specifications ((Figure S2.10).

```
resid_df_1 <- data.frame(date = dat$date, resid = resid(gam_cr1),
                        Model = "s(Phos)")
resid_df_2 <- data.frame(date = dat$date, resid = resid(gam_cr2),
                        Model = "s(Phos, by = date)")
resid_df_3 <- data.frame(date = dat$date, resid = resid(gam_cr3),
                        Model = "s(date, by=Phos)")
resid_df_4 <- data.frame(date = dat$date, resid = resid(gam_cr4),
                        Model = "s(date)")
resid_df_5 <- data.frame(date = dat$date, resid = resid(gam_cr5),
                        Model = "te(Phos, date)")
resid <- rbind(resid_df_1, resid_df_2, resid_df_3, resid_df_4, resid_df_5)
gam_resid <- ggplot(resid, aes(date, resid, group = Model)) +
  geom_point() +
  geom_smooth() + theme_classic() +
```

```
facet_wrap(~ Model) + xlab("Date") +  
ylab("Residuals") + ggtitle("GAM Residuals")
```

There are a few more analysis approaches for running GAMs including the choice of smoother function (Figure S2.8 & Figure S2.11) and the number of knots, or the “wiggleness” (Figure S2.12) of the smoothed relationship. We can plot these different results and include the results of the DLM for comparison, note that ‘cr’ and ‘tp’ are so similar they are difficult to discern from the plot. We can do a similar comparison for the choice of k . Here we compare a four different numbers of knots (Figure S2.12), we show an intentionally wide range of k (30 - 120) to illustrate the effects of k on estimated uncertainty, in practice k should be specified based on degrees of freedom and expectations of variability through time.

Figures

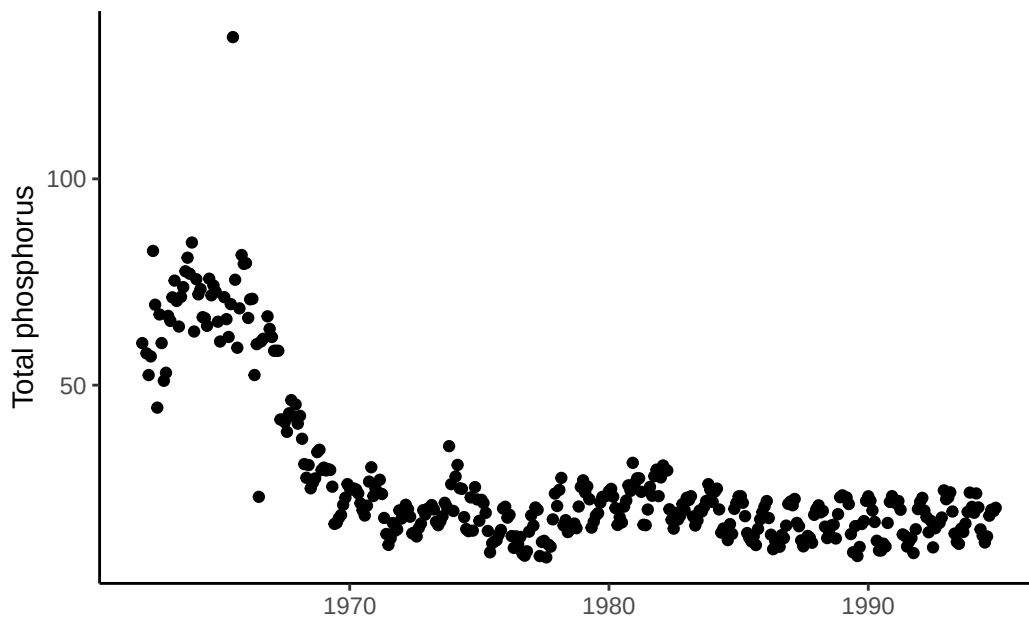


Figure S2.1: Time series of total phosphorus in Lake Washington

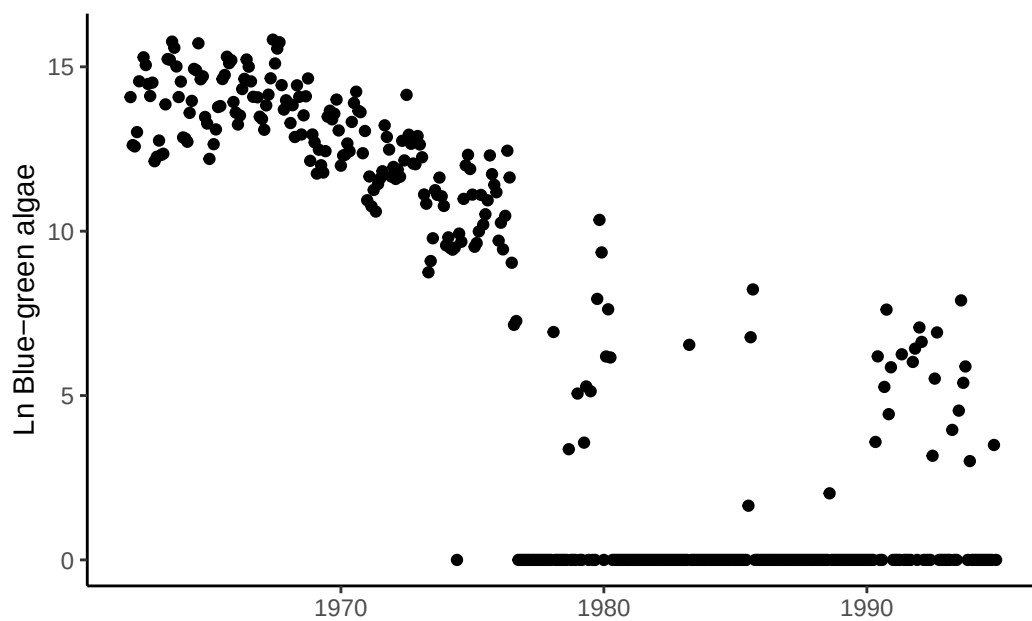


Figure S2.2: Time series of bluegreen algae (cyanobacteria) abundance in Lake Washington

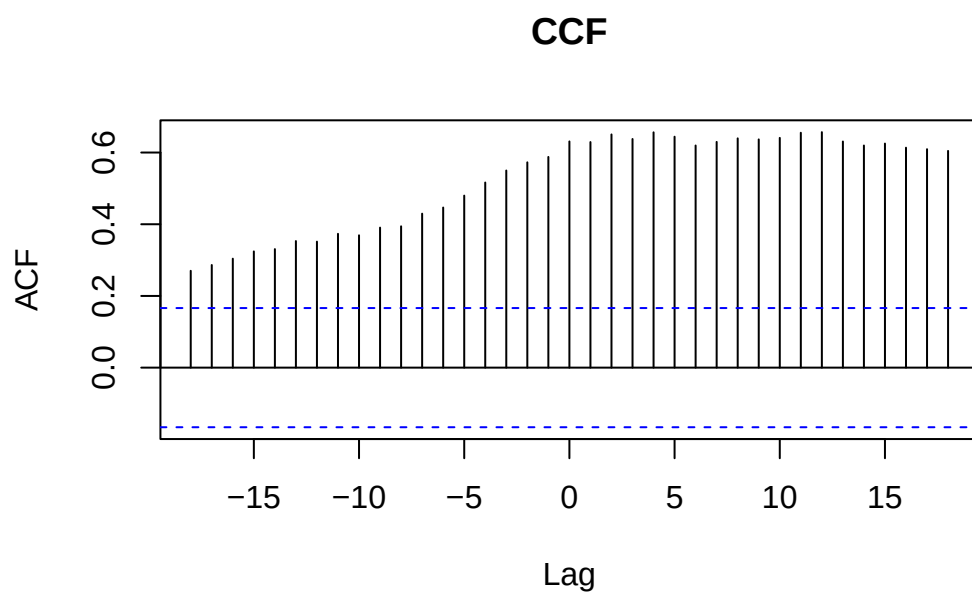


Figure S2.3: A CCF (Cross-Correlation Function) plot that represents the correlation between total phosphorus and bluegreen algae abundance as a function of the lag (time difference) between them.

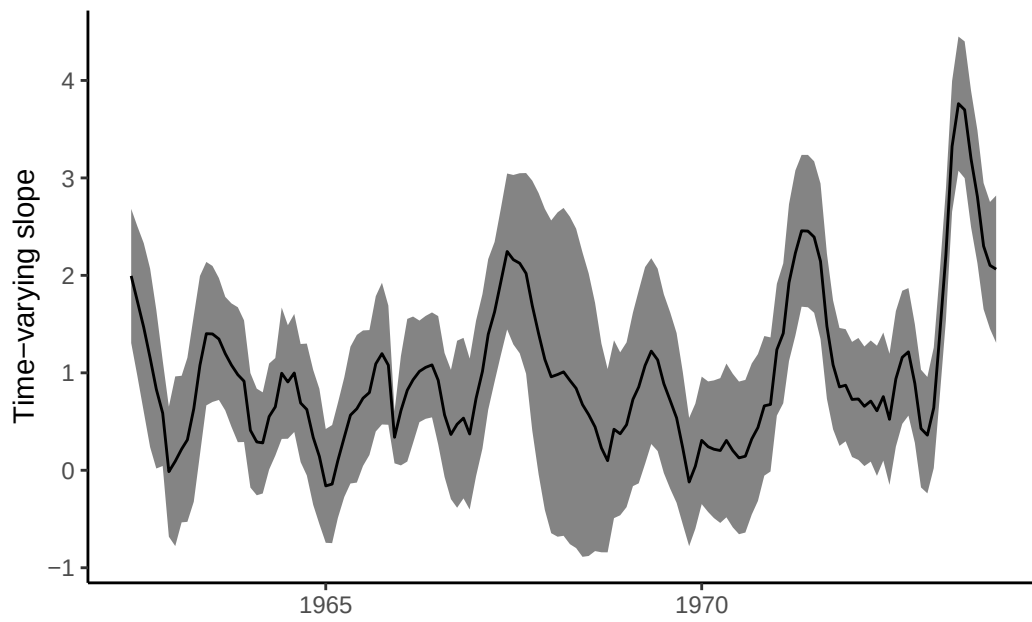


Figure S2.4: Time-varying slope of the relationship between bluegreen algae (response) and total phosphorus (driver) from a DLM

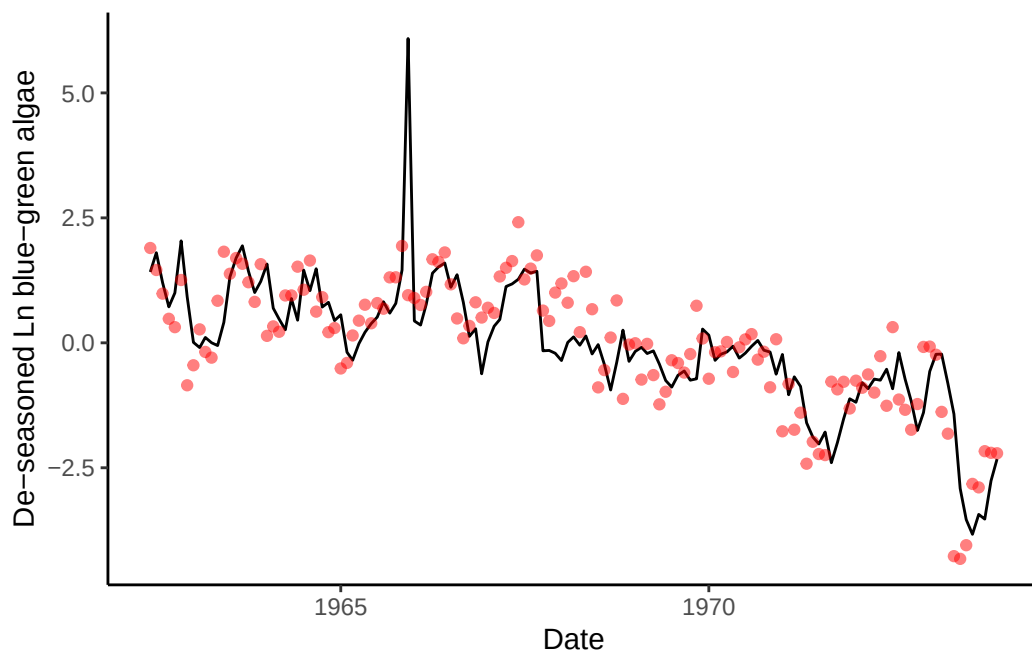


Figure S2.5: Fit of the DLM model with time-varying slope (black line) to observed bluegreen algae abundance (red points).

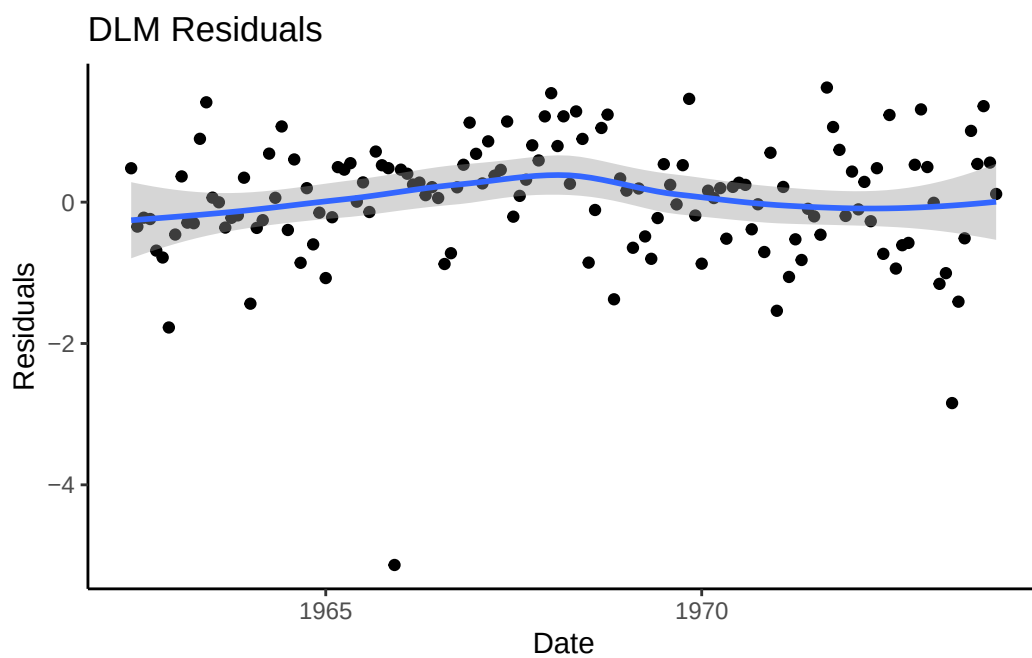


Figure S2.6: Time series of residuals from the DLM model fit with a time-varying slope between blue-green algae and phosphorus

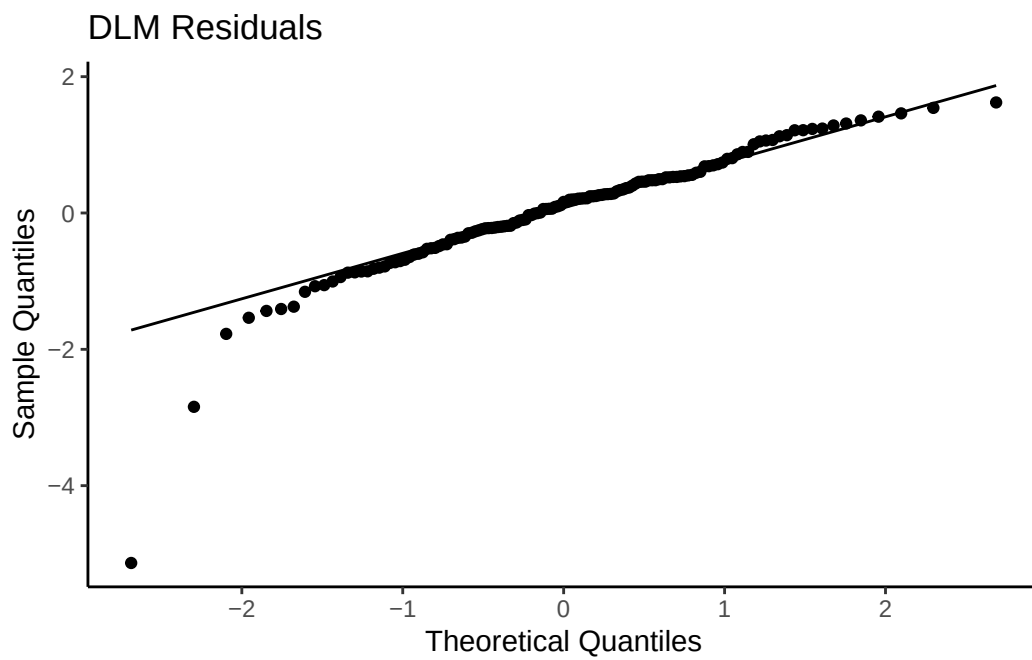


Figure S2.7: Q-Q plot from the DLM model fit with a time-varying slope between blue-green algae and phosphorus

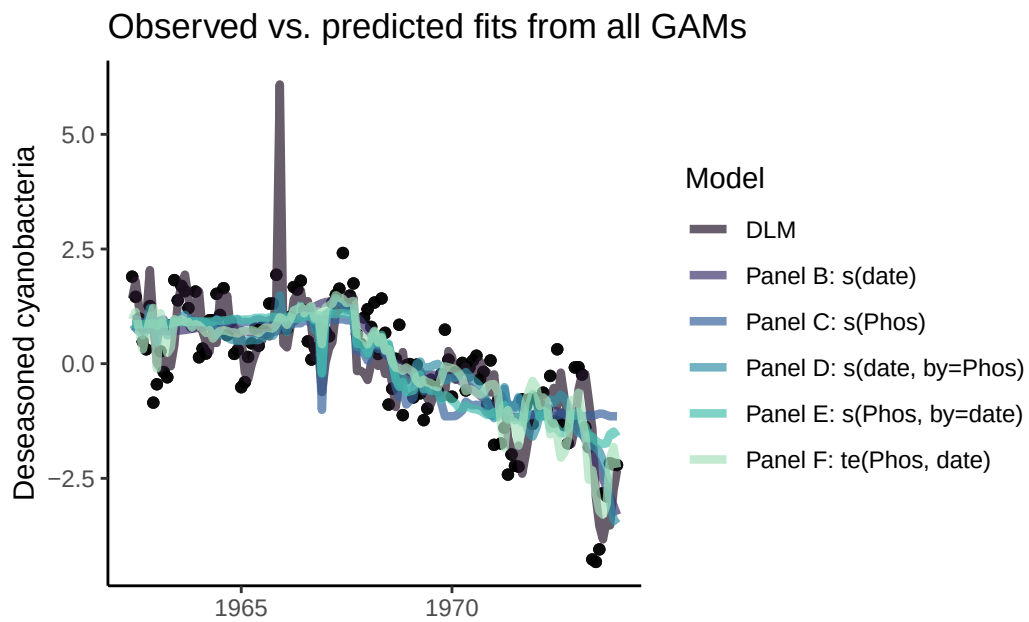


Figure S2.8: Fit of the GAM models (lines) to observed bluegreen algae (cyanobacteria) abundance (black points). Colors denote different specifications of the smoother functions.

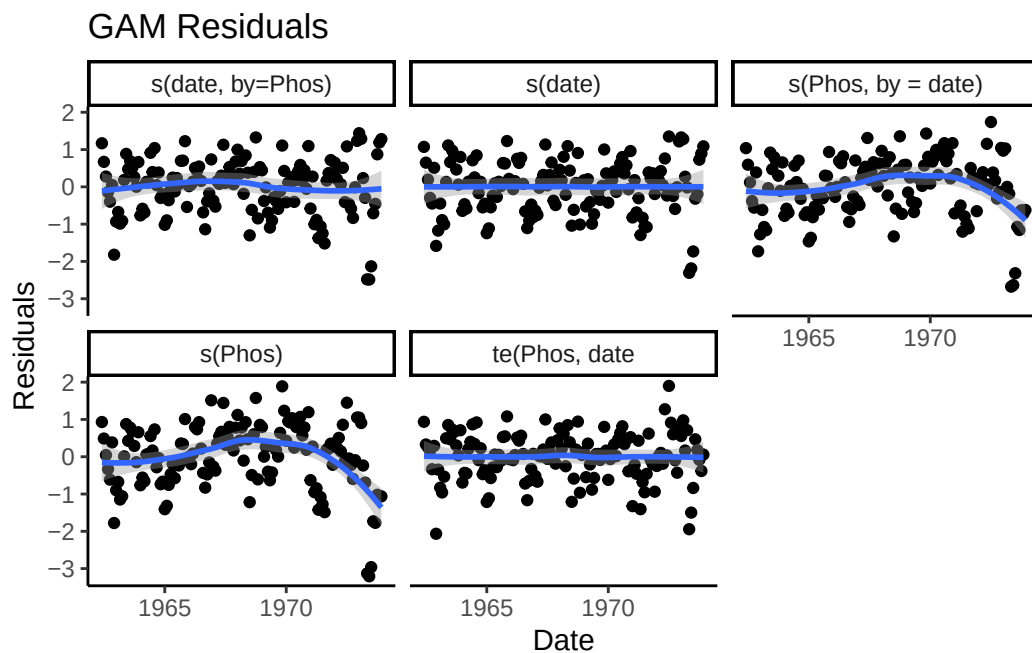


Figure S2.9: Time series of residuals from the five GAM models where residuals from models that do not have a temporally structured smooth term ($s(\text{Phos})$ and $s(\text{Phos}, \text{by} = \text{date})$) have a trend through time.

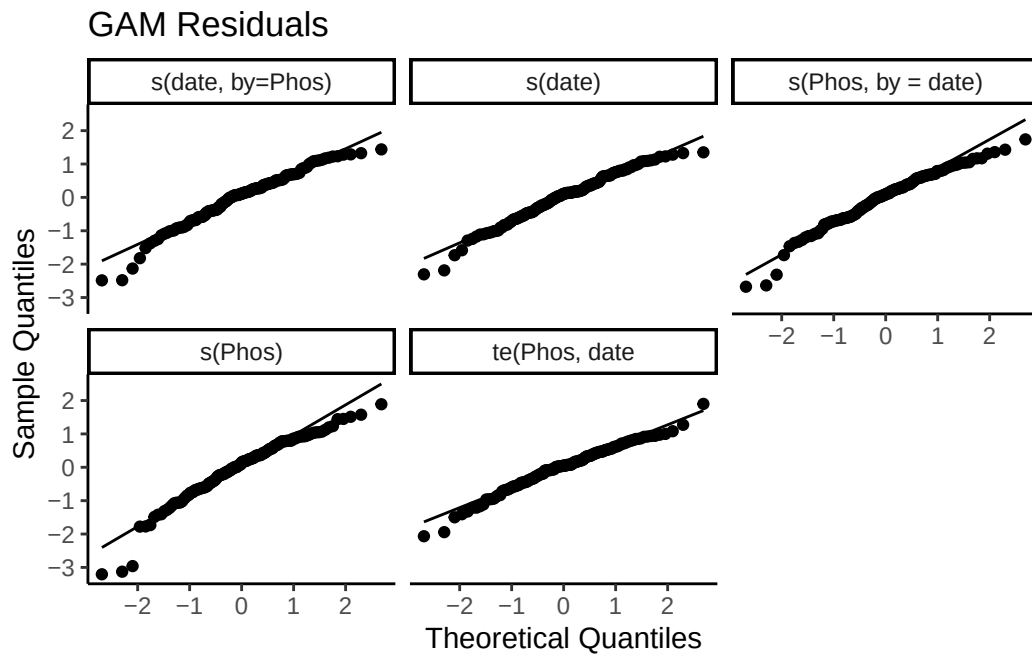


Figure S2.10: Q-Q for each of the five GAM models where residuals from models that do not have a temporally structured smooth term ($s(\text{Phos})$ and $s(\text{Phos}, \text{by} = \text{date})$) have a trend through time.

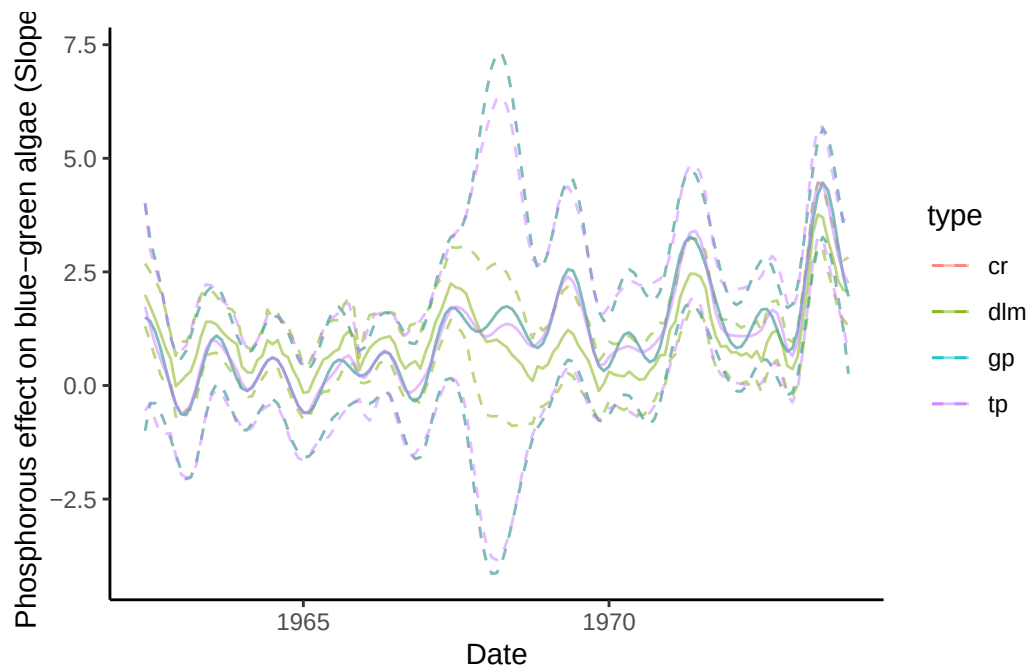


Figure S2.11: Effect of the choice of smoother function on estimates of time-varying slope parameter where color denotes model type. 3 GAM smoother types (cr, tp, gp) are contrasted with a DLM (dlm).

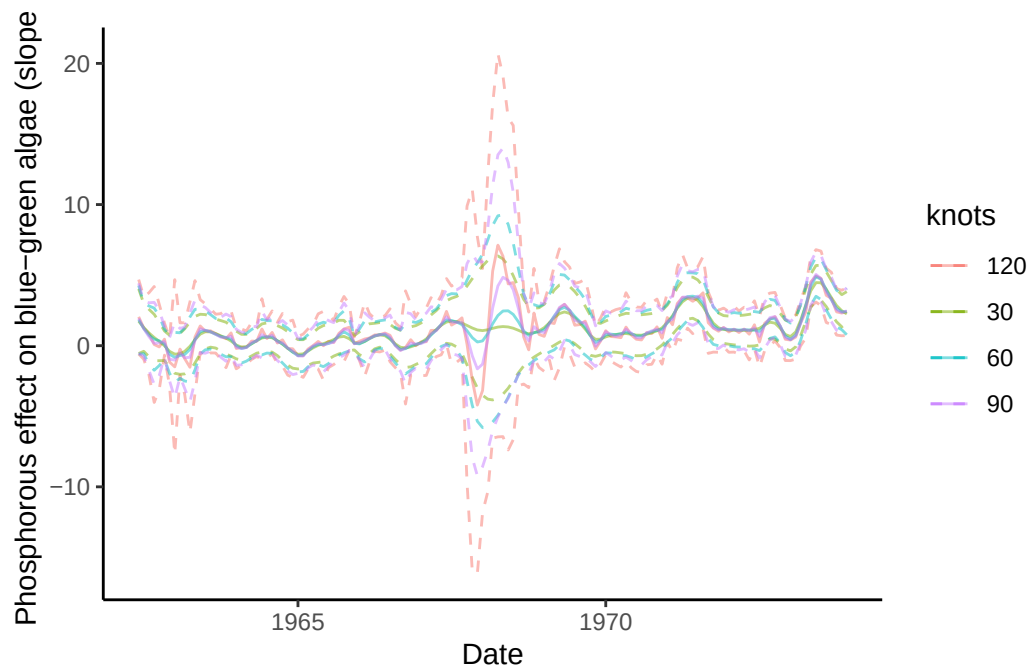


Figure S2.12: Effect of smoothing degrees of freedom on the standard errors of the estimates of time-varying slope parameter where color denotes model type. We show 4 distinct knots (colors)